



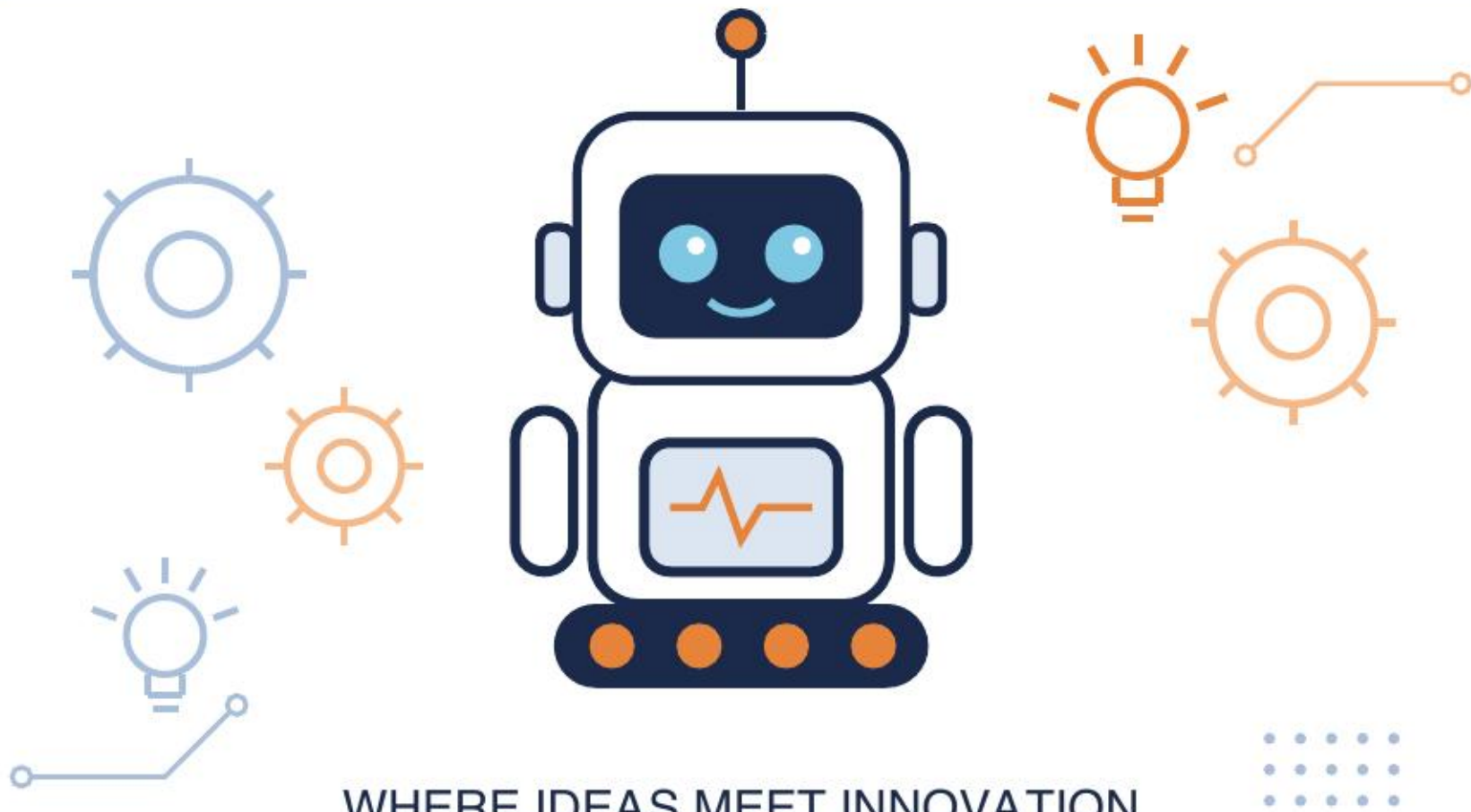
— WELCOME TO OUR —

AI & ROBOTICS

L A B

THINK • BUILD • CODE • LEARN

A calm space to think, tinker and learn



WHERE IDEAS MEET INNOVATION



THINK
ask why



BUILD
make it real



CODE
teach machines



LEARN
grow every day

EVERYONE IS A BEGINNER AT SOMETHING.



OUR LAB CODE

Six habits that make great builders



BE CURIOUS

Ask questions. Wonder how things work.



START SMALL

Simple first. Fancy later.



SHARE THE SPACE

Take turns with kits, tools and computers.



LEARN FROM FAILURE

A robot that fails is teaching you something.



RESPECT EQUIPMENT

Handle gently. Return everything to its place.

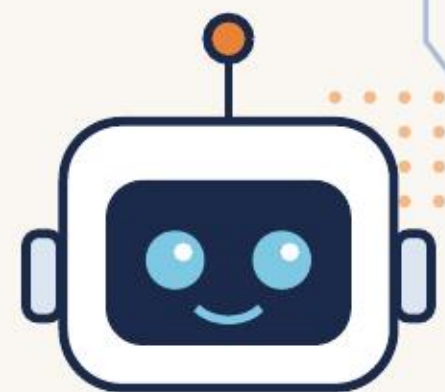


HELP EACH OTHER

Explaining is the best way to learn.



GREAT ROBOTS ARE
BUILT WITH
GREAT HABITS!

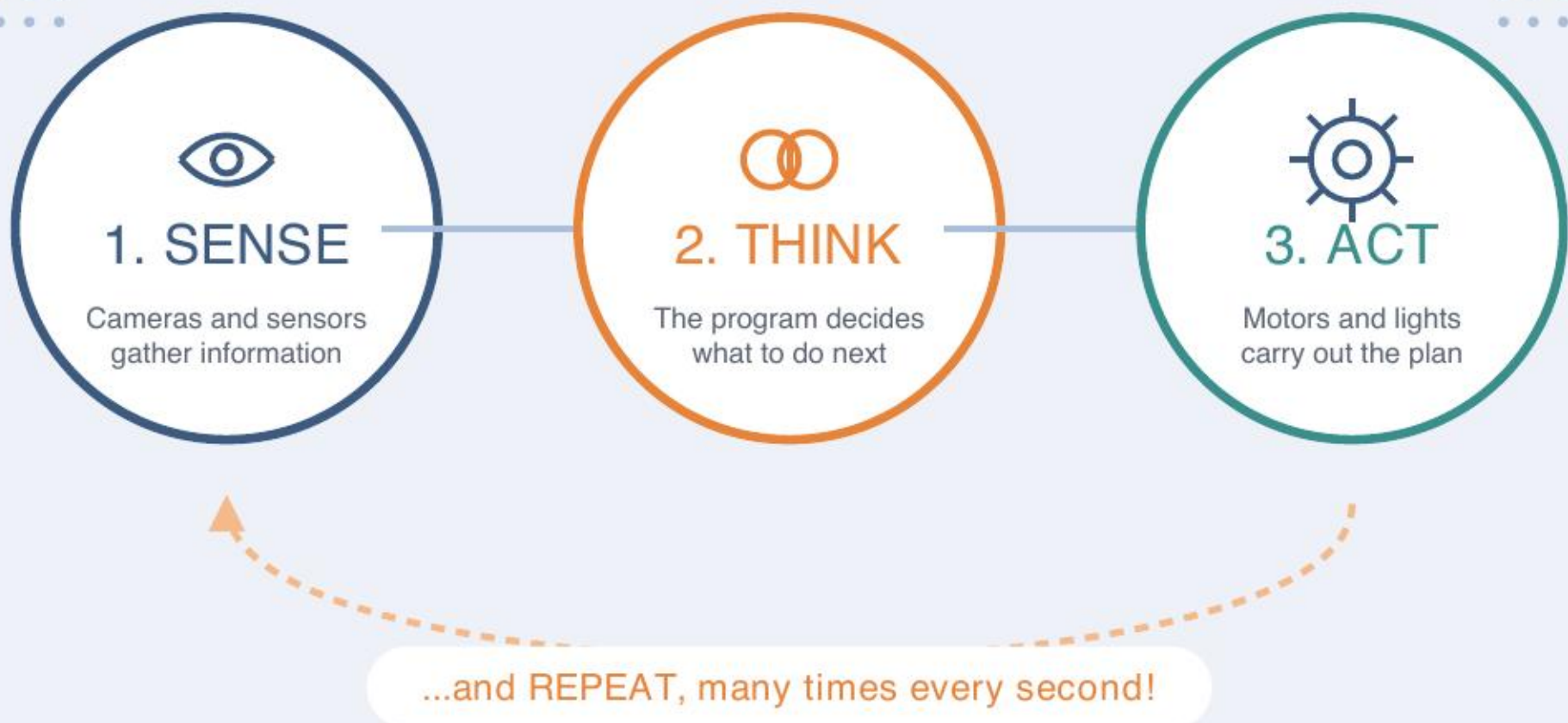


BUILD HABITS. BUILD ROBOTS. BUILD YOURSELF.



HOW DOES A ROBOT THINK?

Every smart machine follows the same simple loop



Your phone, a self-driving car and our lab robots all run this loop.

When you code, **YOU** write the THINK step!



THE DESIGN CYCLE

How engineers turn ideas into inventions



IMAGINE
dream big



DESIGN
sketch it



BUILD
make it



TEST
check it



IMPROVE
level up

IT'S NOT FAILING — IT'S FINDING VERSION 2.



USE AI WISELY

AI is a powerful tool. YOU are still the thinker.

1



THINK FIRST

Try the problem yourself before asking AI.

2



CHECK THE ANSWER

AI can be confidently wrong. Verify facts.

3



BE HONEST

Say when AI helped with your work.

4



PROTECT PRIVACY

Never share personal details with AI tools.

5



STAY KIND

Never use AI to trick or hurt anyone.

SMART TOOLS NEED
WISE HUMANS.

That's you. Use it well.



YOUR BRAIN LEADS. AI ASSISTS.



EVERY BUG IS A CLUE.

Debugging is detective work for builders

"It doesn't work"



"It doesn't work YET"

"I broke it"



"I found a weak spot"

"This is too hard"

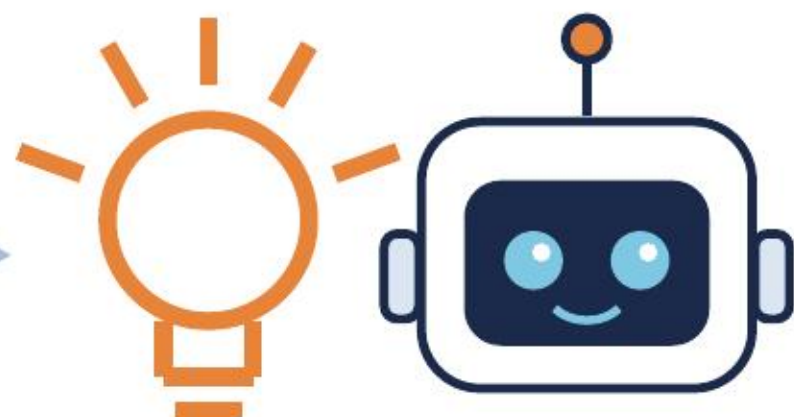


"I need a smaller first step"

"I give up"



"I'll try a different way"



Stuck is just the step before solved.

MISTAKES ARE DATA. COLLECT THEM PROUDLY.



GOOD QUESTIONS BUILD GREAT MINDS

Questions worth asking in this lab

?

What happens if I change this?

?

Why did it do that?

?

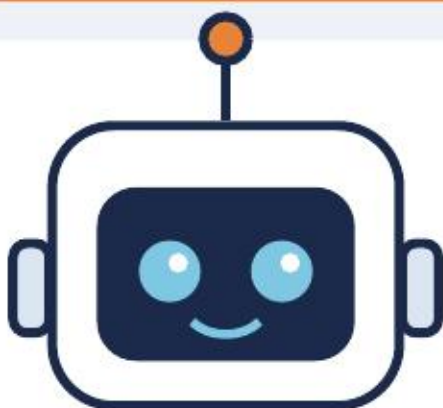
How could this be simpler?

?

Who could this invention help?

?

What would I try next time?



Curiosity is the first tool
in every kit.



OBSERVE · ASK · EXPLORE · DISCOVER

STAY CURIOUS. THE ANSWERS WILL FOLLOW.



SAFE HANDS, SMART BUILDS.

Look after yourself, your team and your tools



POWER DOWN FIRST

Unplug before changing wires or parts.

1



TOOLS HAVE HOMES

Pick up, use, return. Every time.

2



KEEP IT TIDY

A clear table prevents most accidents.

3



ASK BEFORE YOU ACT

Unsure about a tool? Ask a teacher.

4



REPORT BREAKAGES

Broken parts are fine. Hidden ones aren't.

5



A safe lab is a lab where
everyone gets to build.

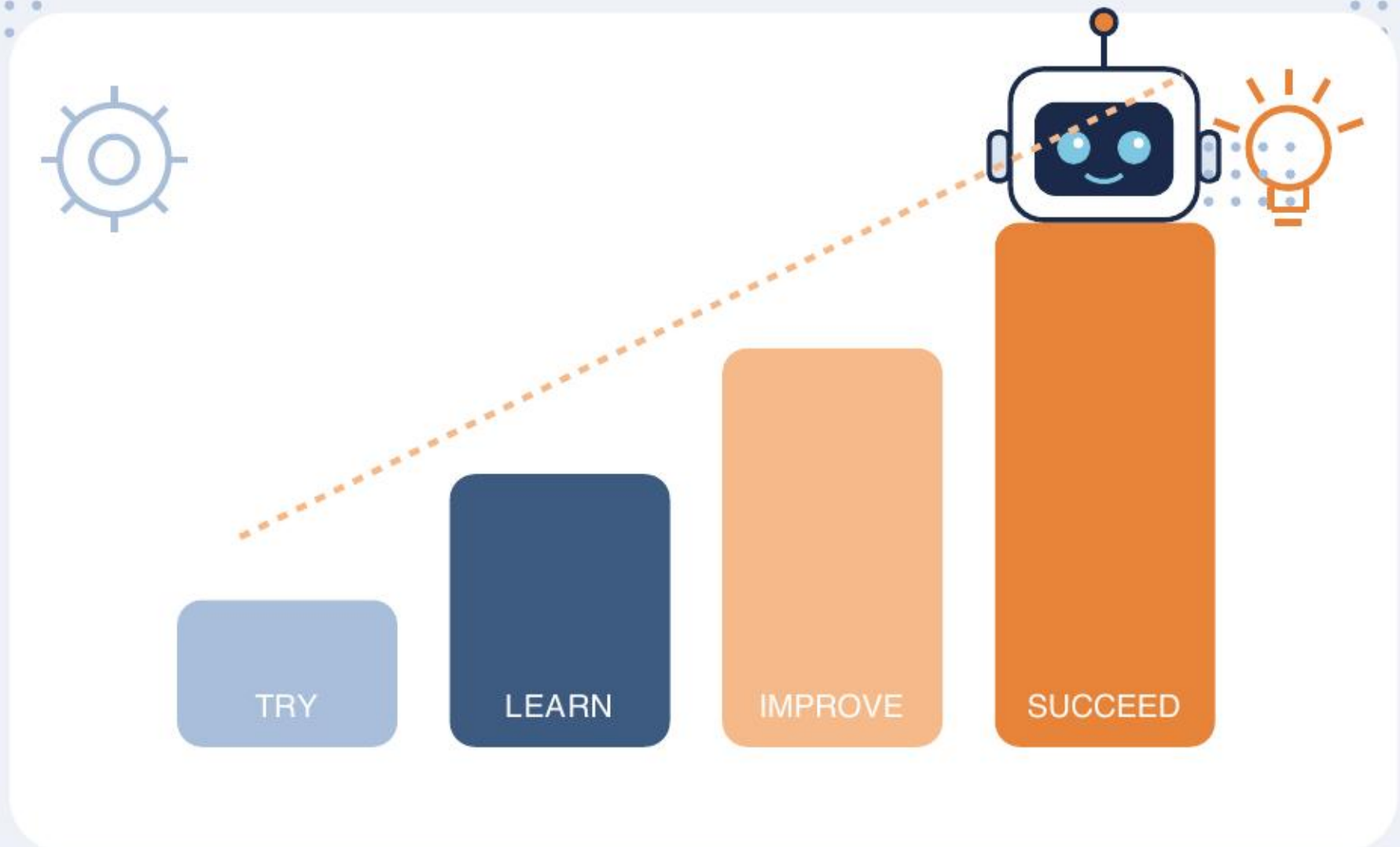


SAFETY IS A TEAM PROJECT.



SMALL STEPS. BOLD IDEAS.

Every invention in history started as a rough first try



DREAM
big



PRACTISE
daily



IMPROVE
together



ACHIEVE
proudly

TODAY'S LEARNERS. TOMORROW'S INVENTORS.



NO ROBOT IS BUILT ALONE.

Great teams share ideas, tools and credit

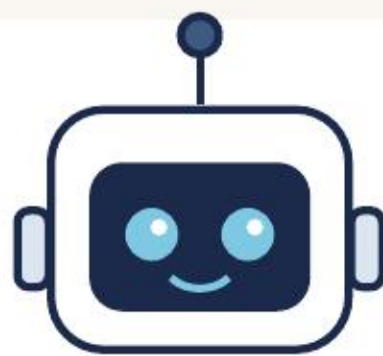
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DESIGNER
sketches the idea

⚙️
BUILDER
assembles the parts

∞
CODER
writes the brain

✓
TESTER
finds what breaks

...and every role swaps around, so everyone learns everything



SHARE IDEAS · SHARE TOOLS · SHARE CREDIT

TOGETHER WE BUILD TOMORROW.



KNOW YOUR ROBOT

Every part has a job — just like your body

SENSORS

the eyes and ears

CONTROLLER

the brain - runs your code

CHASSIS

the skeleton holding it all

WIRES

the nerves carrying signals

BATTERY

the energy source

MOTORS

the muscles that move it



SEE
sensors



THINK
controller



MOVE
motors



POWER
battery

HARDWARE + SOFTWARE = ONE TEAM.



SIMPLE MACHINES, MIGHTY IDEAS.

Six ancient inventions hiding inside every modern robot



LEVER

1

A see-saw for force. Lift heavy things with less effort.



WHEEL & AXLE

2

Rolls friction away. Every robot drive uses one.



PULLEY

3

Change the direction of a pull. Cranes love these.



INCLINED PLANE

4

A ramp trades distance for easier lifting.



WEDGE

5

Concentrates force to split or grip things.



SCREW

6

A ramp wrapped around a rod. Holds builds together.



SPOT THEM
in your builds



COMBINE THEM
into machines



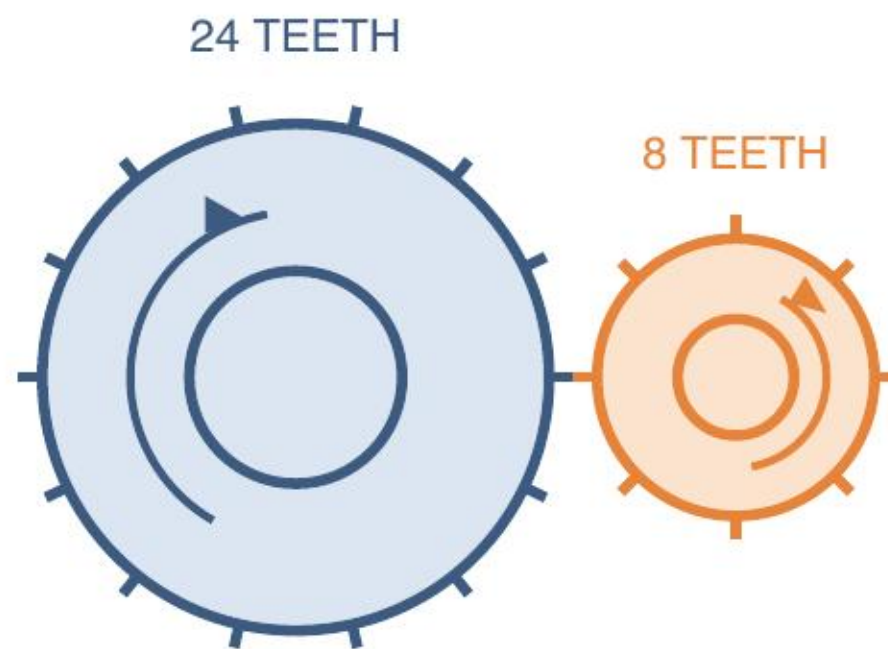
MASTER THEM
become unstoppable

OLD IDEAS. NEW ROBOTS.



THE MAGIC OF GEARS

Trade speed for strength — or strength for speed



One turn of the BIG gear = three turns of the SMALL gear

Count the teeth to predict the trade!

BIG drives SMALL

The small gear spins **FASTER**
but with less force. Racing!



SPEED

small output gear



STRENGTH

big output gear



BALANCE

choose wisely

SMALL drives BIG

The big gear turns **STRONGER**
but more slowly. Climbing!

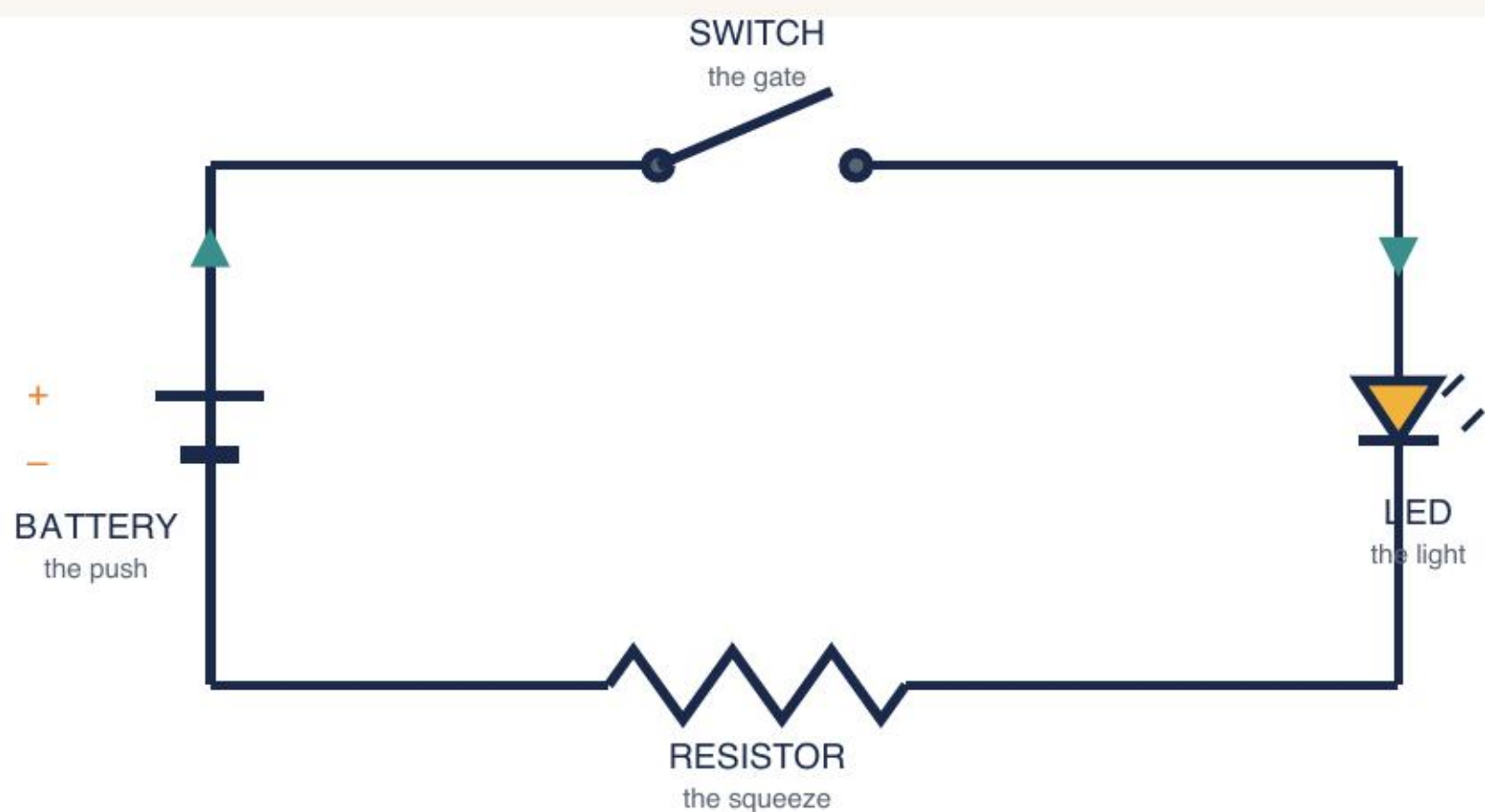


GEAR UP. GEAR DOWN. GEAR SMART.



ELECTRICITY IS A LOOP

No closed loop, no light — every circuit must come home



VOLTAGE

the PUSH
like a water pump



CURRENT

the FLOW
electrons on the move



RESISTANCE

the SQUEEZE
a narrow pipe slows flow



PUSH
voltage



FLOW
current



SQUEEZE
resistance



LOOP
closed circuit

CLOSE THE LOOP. LIGHT THE IDEA.



WIRING WISDOM

Small habits keep circuits (and fingers) happy



RED IS +, BLACK IS –

Colour-code every connection you make.



CHECK POLARITY TWICE

Backwards parts are broken parts.



NEVER SHORT A BATTERY

+ straight to – makes heat. Heat makes smoke.



POWER OFF TO REWIRE

Change the wires first, switch on second.



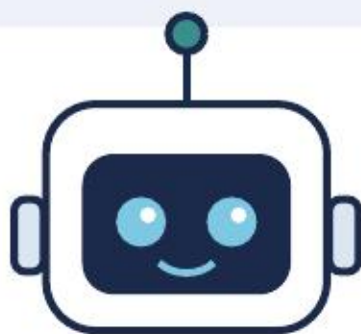
TIDY WIRES SAVE HOURS

Label, bundle and route them neatly.



WARM PART? STOP.

Heat is a warning. Unplug and ask why.



Neat wiring isn't just pretty —
it's how pros find bugs fast.



TOOLBOX HEROES

The right tool makes every build easier — and safer



SCREWDRIVER

Turns screws. Match the tip to the head.



PLIERS

Grips, bends and holds small parts steady.



SPANNER

Tightens nuts and bolts. Snug, not brutal.



MULTIMETER

Measures volts and checks connections.



WIRE STRIPPER

Removes insulation without cutting the wire.



ALLEN KEY

Drives hex bolts in kits and 3D printers.



PICK UP
the right tool



USE
with care



RETURN
to its home

TOOLS DOWN, TABLE CLEAR, HIGH FIVE.